

Batch 1 (SecA): Number system + divisibility + HCF/LCM + fractions entry

CONCEPT	FORMULA / RULE	TRAP
Number sets	N in W in Z in Q; p/q with $q \neq 0$ = rational; else irrational (π , $\sqrt{2}$).	$\sqrt{4} = 2$ rational; $22/7$ only approximates π ; 0 is whole, not natural.
Primes	Prime = 2 factors; test divisibility up to $\sqrt{n}$; 1 is neither; co-prime = gcd 1.	1 not prime; 2 is prime; $91 = 7 \times 13$; $51 = 3 \times 17$.
Parity	$E + E = E$, $O + O = E$, $E + O = O$; $E \times \text{any} = E$; $O \times O = O$.	odd+odd+even = even; track last digit only.
Place vs face	Place = digit $\times$ position; face = digit itself.	7 in 876: place 70, face 7; 0 adds place, face 0.
Factors/multiples	a divides b if $b = a \times k$; LCM of factors lists via tables.	factors(12) = 1,2,3,4,6,12; multiples(6) = 6,12,18...
Remainder	Dividend = divisor $\times$ quotient + r , with $0 \leq r < d$.	$r \geq d$ invalid; verify: $47/6$ gives $q7 \ r5$ since $6 \times 7 + 5 = 47$.
Div 2/3/4	2: ends 0/2/4/6/8. 3: digit-sum divisible by 3. 4: last-two divisible by 4.	Whole-number-divisible-by-3 is false; div-3/4 rules differ.
Div 5/6	5: ends 0/5. 6: BOTH even AND digit-sum divisible by 3.	Needs both; either alone fails.
Div 8/9	8: last-three divisible by 8. 9: digit-sum divisible by 9.	Use last-three, not last-two; div-9 stricter than div-3.
Div 10/11	10: ends 0. 11: (odd-place sum) minus (even-place sum) divisible by 11.	$918082: (9+8+8)-(1+0+2) = 22$, divisible.
Missing digit	Star + rule + test digits 0-9; also least add = next-multiple minus N.	First digit cannot be 0; test, never guess.
HCF/LCM	$a \times b = H \times L$ (two numbers). HCF = division ladder last divisor; words: greatest/divide = HCF, least/together = LCM.	Product rule fails for 3+ numbers; bells 12/18/20 give LCM 180 min = 3 h.

60-SECOND SPEED ROUND (cover answers, recall aloud)

Q: Prime? 1, 2, 51, 97	A: 2, 97 prime; 1 neither; $51 = 3 \times 17$
Q: Div-11: 2728?	A: $(2+2)$ minus $(7+8) = -11$, divisible
Q: H8 L60, a20, find b?	A: $b = 8 \times 60 / 20 = 24$

Batch 2 (SecB): Fractions/decimals + BODMAS + powers + ratio + percent

CONCEPT	FORMULA / RULE	TRAP
Fraction ops	$a/b + c/d = (ad+bc)/bd$; multiply straight, divide by reciprocal.	Add-tops is wrong; always reduce; of = multiply.
Decimal ops	Align points for +/-, count places for x ($1.2 \times 0.4 = 0.48$).	Misaligned points; wrong place count.
Percent bank	$1/2=50$, $1/4=25$, $1/8=12.5$, $1/3=33.33$, $2/3=66.67$, $1/6=16.67$, $1/7=14.28$, $1/9=11.11$, $1/11=9.09$; $37.5\% = 3/8$, $62.5\% = 5/8$.	37.5% is $3/8$, not 3.75 ; memorize bank for speed.
Compare	Cross-multiply ad vs bc; benchmarks $1/2$, $1/4$, $3/4$.	Bigger denominator is NOT bigger.
Terminating	Terminates only if denominator (reduced) = 2^s and 5^s only.	$5/12$ recurs (factor 3); $7/20$ terminates.
BODMAS	Brackets, Orders, Divide/Multiply left-to-right, Add/Subtract left-to-right.	Never add before multiply: $12+8 \times 2-5 = 23$.
Brackets	Innermost first; vinculum and of before outer.	Outer-first fails; of = multiply.
Approximation	Round + compatible numbers + 10% anchor (49×21 approx 1000).	Approximation is legal in DI; do not over-round.
Exponents	Add/subtract/multiply powers by rules; $a^0 = 1$; $(2^3)(2^4) = 2^7 = 128$.	Power-add only for same base; $a^0 = 1$, not 0.
Squares/surds	Memorize squares 1-30, cubes 1-12; $\sqrt{50} = 5\sqrt{2}$; rationalize $1/\sqrt{2} = \sqrt{2}/2$.	$\sqrt{a+b}$ never splits; unrationalized denominator loses marks.
Ratio/proportion	$a:b$ = parts, total = sum; $a/b = c/d$ gives $ad = bc$; share = total $\times$ part/sum.	Ratio is not a value; 550 in 3:5:3 gives 150,250,150.
Percent change	$p\%$ of $N = pN/100$; change% = diff/original $\times 100$; successive net = $a+b+ab/100$ with signs; reverse: $O = F \times 100/(100+r)$.	Never add percents; +20 then -10 nets +8, not +10.

60-SECOND SPEED ROUND (cover answers, recall aloud)

Q: 62.5% as fraction? Bigger 4/7 or 5/9?	A: 5/8; cross 36 vs 35, 4/7 wins
Q: 20 minus $4 \times 3 + 2$?	A: $20 - 12 + 2 = 10$
Q: Net +10 +10? Original after -20% = 320?	A: +21%; 400

Batch 3 (SecC): Points vs % + averages + P&L; + SI/CI + partnership

CONCEPT	FORMULA / RULE	TRAP
Points vs %	Points = new% minus old% (40% to 46% = +6 points, which is +15%).	+6 points is not +6%.
Population %	New = old x (1 +- r/100); 50k +12% = 56k.	Apply on correct base year each step.
Average	avg = sum/n; missing = avg x n minus known sum.	Average is not the middle value.
Combined/weighted	$(n_1a_1+n_2a_2)/(n_1+n_2)$; weights version $\text{sum}(wx)/\text{sum}(w)$.	Never average the averages; 30@40 + 20@55 = 46.
Avg speed	Equal distance: $2xy/(x+y)$; 40+60 gives 48, not 50.	Arithmetic mean of speeds is wrong.
CP/SP/MP	$P\% = (SP-CP)/CP \times 100$; $D\% = (MP-SP)/MP \times 100$.	P&L; always on CP; discount always on MP.
Successive disc	Net = $a+b+ab/100$ with signs; -20 then -10 nets -28%.	Never add discounts.
MP pricing	Fix SP from CP target first, then $\text{discount}\% = (MP-SP)/MP$.	Margin is on CP, discount on MP.
False weight	$\text{Gain}\% = (\text{claim}-\text{actual})/\text{actual} \times 100$; 1000 vs 950 gives 5.26%.	Gain on actual, not claimed.
SI + P/R/T	$SI = PRT/100$; $A = P+SI$; P/R/T from $SI \times 100/RT$; doubling: $R = 100/\text{years}$.	Double in 8y gives $R = 12.5\%$.
CI	$A = P(1+r/100)^n$; half-yearly: $r/2$, $2n$ periods.	Half-yearly is NOT half the SI; adjust then power.
CI-SI + partners	2-yr diff = $P \times R^2/100^2$; growth/decay $P(1+-r/100)^n$; profit ratio C1T1:C2T2; salary-first, split rest.	Capital x months, never capital alone; salary before split.

60-SECOND SPEED ROUND (cover answers, recall aloud)

Q: 45% to 54%?	A: +9 points, which is +20%
Q: CI on 5000 at 8% for 2y?	A: 832 interest (5832 total)
Q: 6000x9 : 9000x6, profit 13000?	A: 1:1 ratio, 6500 each

Batch 4 (SecD): Alligation + work/time + pipes + speed/trains

CONCEPT	FORMULA / RULE	TRAP
Weighted avg	$\text{sum}(wx)/\text{sum}(w)$; e.g. $60 \times 2 + 90 \times 3$ over $5 = 78$.	Weights ignored = wrong mean.
Alligation	$(\text{dear-mean}) : (\text{mean-cheap})$; $12 + 20$ for mean 15 gives 5:3.	Ratio order flips if reversed; verify mean lies between.
Replacement	$\text{Final} = \text{Initial} \times ((T-\text{out})/T)^n$; 100L out 10 twice gives 81L.	Apply power for repeats, not single linear cut.
Ages	Now = x, shift $\pm n$ both sides; difference constant; ratio+sum solves.	Ratio changes after shift; difference never changes.
Work rate	Combined 2 workers: $ab/(a+b)$; $6+12$ days gives 4 days.	Never add days; add rates (reciprocals).
Men-days-hours	$M_1D_1H_1 = M_2D_2H_2$ pairs; alternate-days: 2-day cycle + remainder, note who starts.	Alternate is not half; remainder + starter decide.
Wages	Pay in work ratio = inverse of days ratio.	Equal pay is wrong; faster worker earns more.
Pipes	Net rate = fill minus empty; fill 6h + empty 10h gives net 15h.	Leak is a negative rate; leak-alone from $1/6 - 1/8 = 1/24$, 24h.
Speed + convert	$D = S \times T$; km/h to m/s multiply $5/18$; reverse $18/5$.	km/h = m/s is false; $90 \text{ km/h} = 25 \text{ m/s}$.
Relative/meet	Same direction subtract, opposite add; meet time = dist/sum ; catch-up = dist/diff .	Direction decides add vs subtract.
Avg speed unequal	Total distance/total time; harmonic only for equal legs.	Simple mean of speeds fails on unequal legs.
Trains	Pole: length/speed; platform: $(T+P)/S$; two trains: $(L_1+L_2)/(v_1 \pm v_2)$ with units matched.	Train is not a point; convert km/h before dividing metres.

60-SECOND SPEED ROUND (cover answers, recall aloud)

Q: 80L, out 20 twice, left?	A: $80 \times 0.75^2 = 45\text{L}$
Q: A6d B12d, wages 9000?	A: 6000 and 3000
Q: 200+300m trains, 90 vs 54 opposite?	A: 500m at $40 \text{ m/s} = 12.5 \text{ s}$

Batch 5 (SecE): Boats + clocks + calendar + algebra + 2D mensuration

CONCEPT	FORMULA / RULE	TRAP
Boats	Down = $b+s$, Up = $b-s$; $b = (\text{down}+\text{up})/2$, $s = (\text{down}-\text{up})/2$.	Still-water speed is the mean; stream is half-difference.
Boat times	Time = dist/rate per leg; round trip rarely symmetric.	Upstream always slower; average both legs.
Clock angle	Angle = $ 30H - 5.5M $; hands coincide ~every 65 5/11 min.	5.5 factor, not 6; relative speed 5.5 deg/min.
Gain/loss + leap	Clock gains = extra revolutions; leap year divisible by 4, century by 400.	1900 not leap; 2000 leap.
Odd days	Ordinary 1, leap 2; Sun 0 ... Sat 6; century: 100y=5, 200y=3, 300y=1, 400y=0.	Count odd days mod 7 only.
Calendar repeat	Leap +1... standard repeat cycles 6/11/11; day calc = odd days shift.	Jan-Feb belong to previous year for odd-day counts in leap logic.
Linear equations	One var: isolate; two: elimination/substitution; check by plugging back.	Verify roots; fraction slips hide in transposition.
Identities	$(a+b)^2$, $(a-b)^2$, $a^2-b^2 = (a-b)(a+b)$; use for speed squares.	Expand, never guess; $(a+b)^2$ has +2ab.
Word translation	is = equals; of = multiply; per = divide; integers n, n+1...	Translate fully before solving.
Triangle	Area = $1/2 \times \text{base} \times \text{height}$; Heron $\sqrt{s(s-a)(s-b)(s-c)}$; perimeter sum.	Height must be perpendicular to chosen base.
Circle	$C = 2\pi r$; $A = \pi r^2$; semicircle halves both + diameter on perimeter.	Diameter line counts in semicircle perimeter.
Units	Length $\times 10/100/1000$ chain; area squares it; keep units uniform before computing.	Mixed units = guaranteed error; convert first.

60-SECOND SPEED ROUND (cover answers, recall aloud)

Q: Down 18, Up 10: b? w?	A: b14, w4
Q: Clock angle at 8:24?	A: 240 minus 132 = 108 deg
Q: Leap? 1800, 2000, 2028?	A: No (century/400); yes; yes

Batch 6 (SecEF): 3D mensuration + data interpretation

CONCEPT	FORMULA / RULE	TRAP
Cube	$V = a^3$; $TSA = 6a^2$; diagonal = $a\sqrt{3}$.	Diagonal is $a\sqrt{3}$, not $a\sqrt{2}$.
Cuboid	$V = lbh$; $TSA = 2(lb+bh+hl)$; diagonal = $\sqrt{l^2+b^2+h^2}$.	Three distinct faces; count all pairs.
Cylinder	$V = \pi r^2 h$; $CSA = 2\pi r h$; $TSA = 2\pi r(r+h)$.	CSA vs TSA: lids decide.
Cone	$V = \frac{1}{3}\pi r^2 h$; $l = \sqrt{r^2+h^2}$; $CSA = \pi r l$.	One-third factor; slant height via Pythagoras.
Sphere	$V = \frac{4}{3}\pi r^3$; $A = 4\pi r^2$; hemisphere halves volume, area = $3\pi r^2$.	Hemisphere TSA includes base circle.
Melt/recast	Volume conserved; equate volumes, solve new dimension.	Surface area changes; volume does not.
Capacity	$1\text{ m}^3 = 1000\text{ L}$; $1\text{ cm}^3 = 1\text{ mL}$; convert before comparing vessels.	Units first, arithmetic second.
Table DI	Read headers + units; compute only asked cells; approximate.	Wrong column = wrong set; check units (000, lakh).
Bar DI	Compare heights via ratios; stacked bars split first.	Eyeballing fails close bars; compute ratio.
Pie DI	$\text{Angle}/360 = \text{value}/\text{total}$; $1\% = 3.6\text{ deg}$; combine slices by adding %.	Percent-to-angle factor 3.6.
Line DI	Slope = trend; peaks/troughs = max/min; interpolate linearly only.	Trend is slope, not height.
Mixed + approx	Cross-verify across graphs; round to 2-3 sig figs; 10% anchor.	Approximation legal; over-rounding not.

60-SECOND SPEED ROUND (cover answers, recall aloud)

Q: Cube 9^3 into 3^3 cubes?	A: $729/27 = 27$ cubes
Q: 30% sector of total 5400?	A: 1620
Q: Cylinder $r14\ h20$ (22/7)?	A: 12320